Problem 1: If A and B are orthogonal matrices of the same order, then show that:

(i) AB is orthogonal.

(ii) A^T is orthogonal.

(iii) A^{-1} is orthogonal.

Solution

(i) Since A and B are orthogonal, we have:

$$A^TA = I \quad \text{and} \quad B^TB = I$$

Consider $(AB)^T(AB)$:

$$(AB)^T(AB) = B^TA^TAB = B^TIB = B^TB = I$$

Hence, AB is orthogonal.
(ii) Since A is orthogonal, we know:

$$A^TA = I \Rightarrow AA^T = I$$

This implies:

$$(A^T)^TA^T = AA^T = I$$

So, A^T is orthogonal.
(iii) For an orthogonal matrix A, we know:

$$A^T = A^{-1}$$

But from part (ii), A^T is orthogonal. Therefore, A^{-1} is also orthogonal.

Problem 2: Determine whether the matrix $A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$ is diagonalizable.

Solution

We find the eigenvalues by solving the characteristic equation:

$$\det(A - \lambda I) = \det \begin{bmatrix} -\lambda & 1 \\ -1 & 2-\lambda \end{bmatrix} = (-\lambda)(2-\lambda) - (-1)(1) = \lambda^2 - 2\lambda + 1 = (\lambda - 1)^2$$

So the eigenvalue is $\lambda = 1$ with algebraic multiplicity 2.
Now find the eigenvectors by solving $(A - I)\mathbf{x} = 0$:

$$A - I = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \Rightarrow \text{Row reduce: } \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \Rightarrow x_1 = x_2$$

So the eigenspace is spanned by

$$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \text{dimension 1.}$$

Since the geometric multiplicity (1) is less than the algebraic multiplicity (2), the matrix is **not diagonalizable**.

Problem 3: Determine whether the matrix

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$$

is diagonalizable.

Solution

Find the eigenvalues by solving the characteristic equation:

$$\det(A - \lambda I) = \det \begin{bmatrix} 3-\lambda & -1 \\ 1 & 1-\lambda \end{bmatrix} = (3-\lambda)(1-\lambda) + 1 = \lambda^2 - 4\lambda + 4 = (\lambda - 2)^2$$

So $\lambda = 2$ is the only eigenvalue, with algebraic multiplicity 2.
Now solve $(A - 2I)\mathbf{x} = 0$:

$$A - 2I = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \Rightarrow \text{Row reduce: } \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \Rightarrow x_1 = x_2$$

So the eigenspace is

$$\text{span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}, \quad \text{dimension 1.}$$

Since the geometric multiplicity is less than the algebraic multiplicity, the matrix is **not diagonalizable**.

Problem 4: If matrix R_1 gives a rotation through θ (counterclockwise), and R_2 gives a rotation through ϕ, then what is the geometric significance of R_1R_2?

Solution

The rotation matrix R_1 for a counterclockwise rotation by an angle θ is given by:

$$R_1 = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Similarly, the rotation matrix R_2 for a counterclockwise rotation by an angle ϕ is:

$$R_2 = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$$

Now, perform the matrix multiplication R_1R_2:

$$R_1R_2 = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$$

Multiplying these matrices:

$$R_1R_2 = \begin{bmatrix} \cos\theta\cos\phi - \sin\theta\sin\phi & -\cos\theta\sin\phi - \sin\theta\cos\phi \\ \sin\theta\cos\phi + \cos\theta\sin\phi & -\sin\theta\sin\phi + \cos\theta\cos\phi \end{bmatrix}$$

Using the trigonometric identities:

$$\cos(A+B) = \cos A\cos B - \sin A\sin B, \quad \sin(A+B) = \sin A\cos B + \cos A\sin B$$

We simplify to:

$$R_1R_2 = \begin{bmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) \\ \sin(\theta+\phi) & \cos(\theta+\phi) \end{bmatrix}$$

Thus, R_1R_2 represents a rotation by the angle $\theta + \phi$.

$$\boxed{R_1R_2 = \text{Rotation by } \theta + \phi.}$$

Problem 5: If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, show that the solutions of the characteristic equation of A are represented by: $\lambda = \frac{1}{2}\left[a + d \pm \sqrt{(a-d)^2 + 4bc}\right]$ Also, find the eigenvalues of: $A = \begin{bmatrix} 1 & 2 \\ 1 & -3 \end{bmatrix}$

Solution

$A - \lambda I = \begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix}$, and the determinant is:

$$\det(A - \lambda I) = (a-\lambda)(d-\lambda) - bc = \lambda^2 - (a+d)\lambda + (ad - bc)$$

The characteristic equation is:

$$\lambda^2 - (a+d)\lambda + (ad - bc) = 0$$

This is a quadratic equation, and the solutions for λ are given by the quadratic formula:

$$\lambda = \frac{-(a+d) \pm \sqrt{(a+d)^2 - 4(ad-bc)}}{2}$$

Simplifying further:

$$\lambda = \frac{1}{2}\left[a + d \pm \sqrt{(a-d)^2 + 4bc}\right]$$

Now, we apply this to the given matrix $A = \begin{bmatrix} 1 & 2 \\ 1 & -3 \end{bmatrix}$.
Here, $a = 1$, $b = 2$, $c = 1$, and $d = -3$. Substituting these values into the eigenvalue formula:

$$\lambda = \frac{1}{2}\left[1 + (-3) \pm \sqrt{(1-(-3))^2 + 4(2)(1)}\right] = \frac{1}{2}\left(-2 \pm \sqrt{24}\right)$$

Thus, the eigenvalues for $A = \begin{bmatrix} 1 & 2 \\ 1 & -3 \end{bmatrix}$ are:

$$\lambda_1 = \frac{-2 + 2\sqrt{6}}{2} = -1 + \sqrt{6}$$

$$\lambda_2 = \frac{-2 - 2\sqrt{6}}{2} = -1 - \sqrt{6}$$

Problem 6: Prove that the eigenvectors of A corresponding to different eigenvalues are also distinct.

Solution

Assume for the sake of contradiction that $\mathbf{v_1} = \mathbf{v_2}$ for $\lambda_1 \neq \lambda_2$. Then, from the eigenvalue equations, we would have:

$$A\mathbf{v_1} = \lambda_1\mathbf{v_1} \quad \text{and} \quad A\mathbf{v_2} = \lambda_2\mathbf{v_2}$$

Since $\mathbf{v_1} = \mathbf{v_2}$, we would obtain:

$$A\mathbf{v_1} = A\mathbf{v_2} \quad \Rightarrow \quad \lambda_1\mathbf{v_1} = \lambda_2\mathbf{v_2}$$

This implies:

$$\lambda_1\mathbf{v_1} = \lambda_2\mathbf{v_1}$$

Since $\mathbf{v_1}$ is a non-zero eigenvector (by definition of eigenvectors), we can factor out $\mathbf{v_1}$:

$$(\lambda_1 - \lambda_2)\mathbf{v_1} = 0$$

Now, since $\mathbf{v_1} \neq 0$, it follows that $\lambda_1 - \lambda_2 = 0$, which implies that $\lambda_1 = \lambda_2$.
But this contradicts our assumption that $\lambda_1 \neq \lambda_2$. Therefore, our assumption that $\mathbf{v_1} = \mathbf{v_2}$ must be false.
Thus, the eigenvectors corresponding to distinct eigenvalues are linearly independent, and hence, they must be distinct.

Problem 7: Prove that if λ is an eigenvalue of A, $\mathbf{x}$ is a corresponding eigenvector, and s is a scalar, then $\lambda - s$ is an eigenvalue of $A - sI$, and $\mathbf{x}$ is a corresponding eigenvector.

Solution

By definition of eigenvalue and eigenvector, we have:

$$A\mathbf{x} = \lambda\mathbf{x}$$

Consider the matrix $A - sI$. We will check if $\mathbf{x}$ is an eigenvector of $A - sI$ with eigenvalue $\lambda - s$.
We compute:

$$(A - sI)\mathbf{x} = A\mathbf{x} - sI\mathbf{x}$$

Using the fact that $I\mathbf{x} = \mathbf{x}$, we have:

$$(A - sI)\mathbf{x} = A\mathbf{x} - s\mathbf{x}$$

From the assumption $A\mathbf{x} = \lambda\mathbf{x}$, we substitute:

$$(A - sI)\mathbf{x} = \lambda\mathbf{x} - s\mathbf{x} = (\lambda - s)\mathbf{x}$$

Thus, $\mathbf{x}$ is an eigenvector of $A - sI$ with eigenvalue $\lambda - s$.